

Software Engineering

Lab 1: Requirements Engineering & UML Use-Case Modelling

Problem Statement #36 — Restaurant Table Booking & Pre-Ordering App

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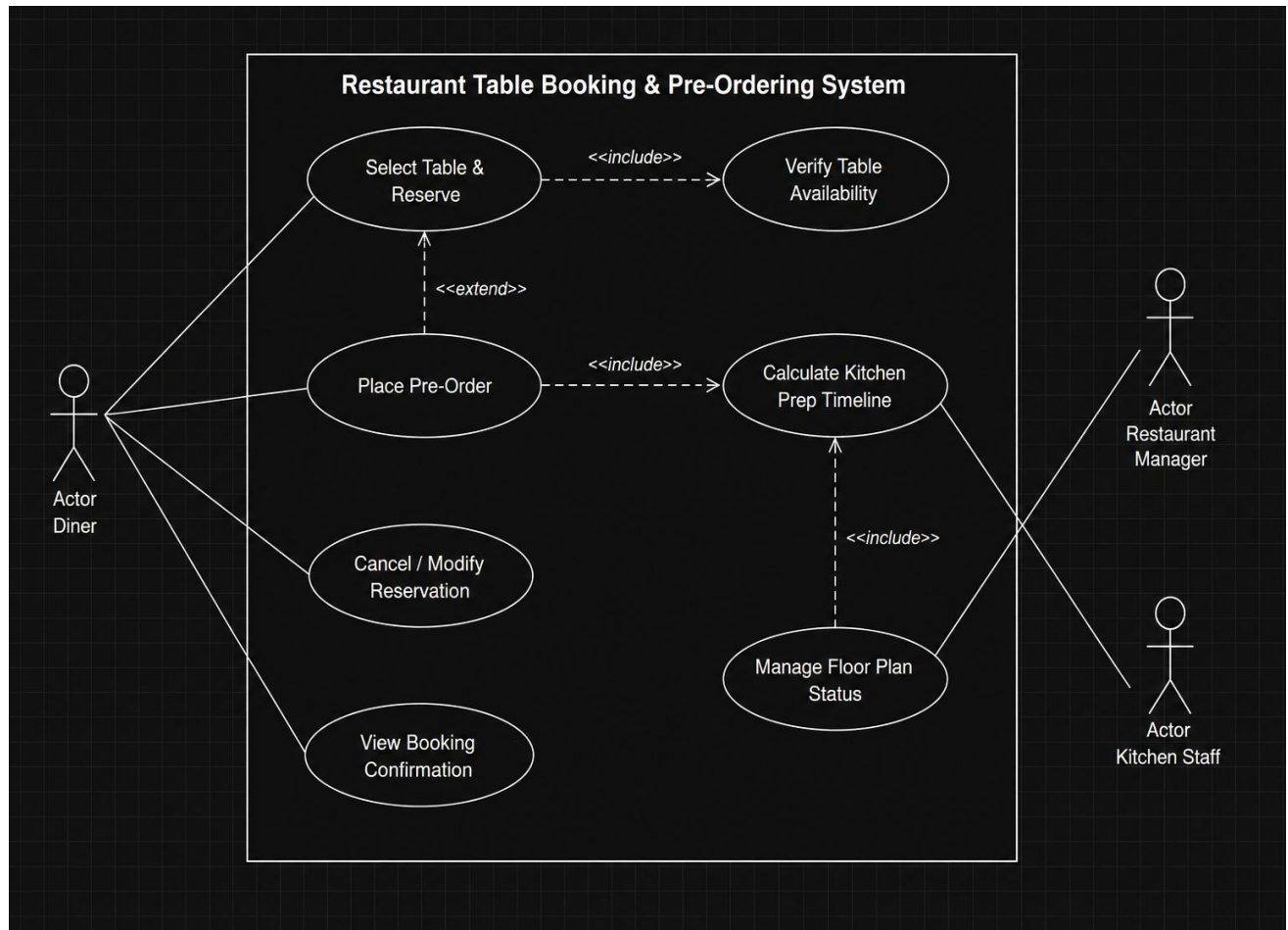
Section: G

1. Requirement Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow diners to pick an open table from an interactive 2D floor plan and link pre-ordered menu items to the reservation.	High	Pass: Table reserved and kitchen pre-order ticket queued. Fail: Overbooking of table during overlapping dining window.	Establishes the core reservation-to-order link that the rest of the app depends on.
FR-002	Functional	The system shall allow diners to browse the restaurant's live menu and add items to a pre-order attached to their table reservation.	High	Pass: Selected item appears in the reservation's pre-order ticket with correct quantity and price. Fail: Item is added but not reflected in the ticket sent to the kitchen.	Let's diners order ahead so food can be prepared before arrival, reducing wait time.
FR-003	Functional	The system shall calculate and display an estimated kitchen preparation timeline whenever a pre-order is confirmed.	High	Pass: Timeline is shown within 2 seconds of confirmation and is accurate to within +/- 5 minutes of actual kitchen completion. Fail: Timeline is missing or off by more than 10 minutes.	Aligns diner arrival time with food readiness for a seamless dine-in experience.
FR-004	Functional	The system shall allow diners to modify or cancel a table reservation up to 30	Medium	Pass: Reservation is updated or cancelled and the table is released for	Improves table turnover by freeing slots that

		minutes before the reserved time.		rebooking. Fail: Cancellation is blocked or the table remains locked after cancellation.	would otherwise sit as no-shows.
FR-005	Functional	The system shall allow the restaurant manager to update a table's status (Reserved, Seated, Billed) from a live floor-plan dashboard.	High	Pass: Status change is saved and visible on the manager dashboard immediately. Fail: Status change fails to save or shows a stale value.	Gives staff an accurate, real-time view of the floor to seat and turn tables efficiently.
NFR-001	Nonfunctional (Performance)	Table status changes (Reserved, Seated, Billed) must sync across all manager floor terminals in under 500 ms.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load. Fail: Sync latency exceeds 500 ms on any terminal during peak load.	Prevents double-seating and staff confusion caused by stale floor-plan data.
NFR-002	Nonfunctional (Security)	The system shall encrypt diner personal and payment data in transit and at rest, and enforce role-based access so only authorized managers can view or edit floor-plan and reservation data.	High	Pass: Penetration test confirms data is encrypted end-to-end and unauthorized roles are denied access. Fail: Any diner or payment field is retrievable unencrypted, or a non-manager role can edit floor data.	Protects sensitive diner and payment information and limits floor-plan control to authorized staff.

2. Use Case Diagram



3. Use-Case Flow Specification

Use Case: Select Table & Reserve (with Place Pre-Order)

Primary Actor: Diner

Preconditions

- The diner has the app open and is logged in.
- The restaurant's live floor plan shows at least one table with status "Open" for the requested time slot.

Postconditions

- The selected table is locked to the diner's reservation and its status changes to "Reserved" on the floor plan.

- A kitchen pre-order ticket is queued and linked to the reservation.
- A confirmation with the reservation time, table number, and estimated kitchen prep timeline is sent to the diner.

Main Success Scenario

1. Diner opens the "Reserve a Table" option and the system displays the live 2D floor plan.
2. Diner selects an open table for a desired date and time.
3. System verifies table availability for that time window (include: Verify Table Availability).
4. System locks the table and prompts the diner to add a pre-order.
5. Diner browses the menu and adds items to the pre-order.
6. System links the pre-order to the reservation and calculates a kitchen preparation timeline (include: Calculate Kitchen Prep Timeline).
7. System displays the reservation and pre-order summary for diner confirmation.
8. Diner confirms the reservation.
9. System updates the table status to "Reserved" on the manager's floor-plan dashboard and queues the kitchen ticket.
10. System sends a confirmation with table number, time, and estimated prep timeline. Use case ends successfully.

Alternate Flow

3a. Table Becomes Unavailable During Selection

3a1. System detects that another diner reserved the same table for an overlapping window before this reservation was confirmed.

3a2. System displays a "Table No Longer Available" message and returns the diner to the floor plan.

3a3. System highlights the next-nearest open tables for the same time slot as suggested alternatives.

3a4. If the diner selects an alternative table, the flow resumes at step 3. If no table is accepted, the reservation attempt is cancelled and the diner is notified.

UML Relationship Notes

«include»: Select Table & Reserve includes Verify Table Availability; Place Pre-Order includes Calculate Kitchen Prep Timeline.

«extend»: Cancel / Modify Reservation extends Select Table & Reserve as optional reservation-management behavior.